

RESEARCH ARTICLE

Development of a Conformational Sampling Method for Transition State Structures With Diverse Active Bond Lengths

Satoshi Matsutani¹  | Taichi Shomura¹  | Nobuya Tsuji²  | Satoshi Maeda^{2,3} ¹Graduate School of Chemical Sciences and Engineering, Hokkaido University, Sapporo, Japan | ²Institute for Chemical Reaction Design and Discovery (WPI-ICReDD), Hokkaido University, Sapporo, Japan | ³Department of Chemistry, Graduate School of Science, Hokkaido University, Sapporo, Japan**Correspondence:** Nobuya Tsuji (tsuji@icredd.hokudai.ac.jp) | Satoshi Maeda (smaeda@eis.hokudai.ac.jp)**Received:** 3 April 2026 | **Revised:** 16 July 2026 | **Accepted:** 25 July 2026

ABSTRACT

We describe TStrail, a transition state (TS) conformational sampling tool that automatically generates TS conformers without fixing the bond forming/breaking distance of an initial TS structure. Conventionally, the conformations of TS structures were searched with the fixed length of active bonds, making it easier to re-optimize to the TS but potentially limiting the comprehensiveness of TS conformers when the active bond lengths varied among conformers. By collectively performing pathway searches among various reactant conformations, TStrail was able to generate TS conformers with different active bond lengths. Furthermore, applications to more complex systems demonstrated that TStrail provides a practical tool for systematically analyzing challenging TS conformers, particularly for large, flexible, ionic, and structurally complex TSs, thereby enabling deeper mechanistic insights through the identification of diverse TS conformers and reaction pathways.

1 | Introduction

Identifying the transition state (TS) on the potential energy surface (PES) is essential for understanding chemical reactions and reaction networks. Recent advancements in automatic TS localization have made TS structure calculations more accessible for many chemists [1–9]. Broadly, automated TS searches can be categorized into two approaches: single-ended and double-ended. A single-ended approach starts with one structure, usually a local minimum on the PES, and employs an algorithm to refine that structure into a first-order saddle point. Examples include the single-ended growing string method (SE-GSM) [10], transition state search using chemical dynamics simulations (TSSCDS) [11], the ab initio nanoreactor [12], imposed activation (IACATA) [13], Chemoton [14], combined molecular dynamics and coordinate driving (MD/CD) method [15], multicoordinate driven (MCD) method [16], pyMCD [17], the artificial force induced reaction (AFIR) method [18], and more. In contrast, a double-ended approach requires two structures, a reactant and a product, to locate the TS. Examples include the string method [19],

nudged elastic band (NEB) [20], double-ended growing string method (DE-GSM) [21], autodE [22], root mean square deviation push-pull (RMSD-PP) [23], ConfoTS [24], synchronous transit and quasi-Newton (STQN) method [25], freezing string method (FSM) [26], and others.

Once a TS with the correct reaction coordinate is located, the next step typically involves finding its lowest-energy conformers. Broadly, this can be approached through two strategies: constrained TS conformational sampling starting from an already identified TS structure, and TS searches starting from multiple reactant/product structures or conformers. The constrained TS conformational sampling typically begins by fixing the distance between atoms involved in bond formation or cleavage, often referred to as the “active bond”, and then performing a conformational search on the rest of the molecule (Figure 1A). This constrained approach can generate many TS-like conformers that retain the same active bond distance as the initial TS. These TS-like geometries are then further optimized to yield unique TS conformers, and the process is repeated until no TS with

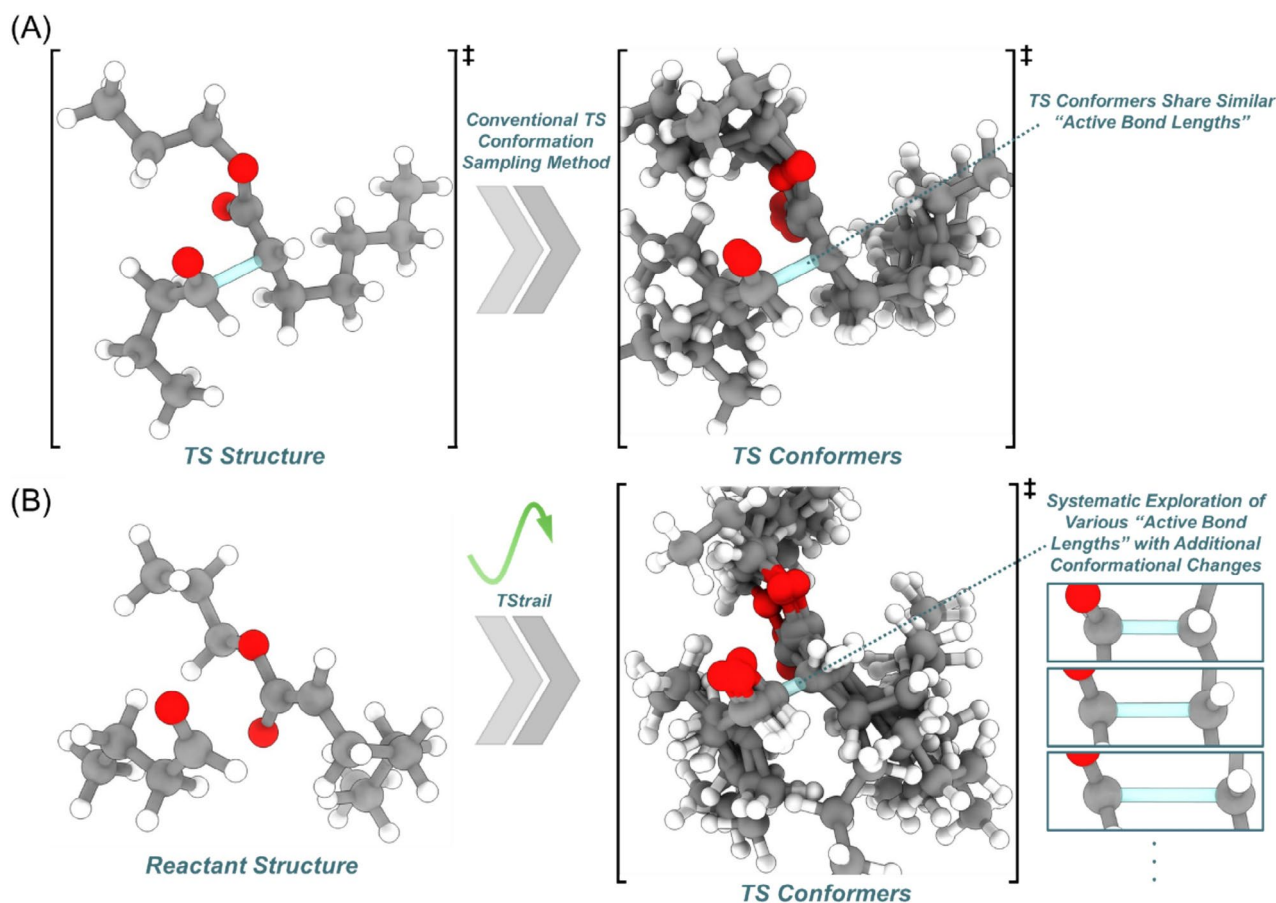


FIGURE 1 | (A) Conventional active bond constrained method for TS conformation sampling. In this example, the bond forming C–C length is fixed, while the remaining parts of the molecule are conformationally sampled. (B) Schematic representation of TStrail. Starting from a reactant structure, TStrail generates TS conformers with various reaction coordinate bond lengths, together with conformational variations in the non-reactive regions.

lower energy is found. Since stereoselectivity in asymmetric reactions is evaluated based on the relative energies of competing TS conformers, identifying the lowest-energy TS conformers through TS conformational sampling is essential for accurate computational analyses [27–29]. One major advantage of constrained TS conformer sampling is its ability to easily generate TS-like conformers that can readily be optimized to a true TS by re-running the TS optimization. Consequently, this active bond constrained method has been incorporated into various TS conformation generation tools, including the Conformer-Rotamer Ensemble Sampling Tool (CREST) [30], An Automated Reaction Optimizer for New catalysts (AARON) [31], QMTSDock [32], and racer^{TS} [33].

Although these procedures have been successfully demonstrated in many reactions, they are based on one critical assumption: all possible TS conformers share the same, or at least very similar, active bond distance. Conducting a conformational search while constraining an active bond distance to its length in the TS will indeed generate many TS-like conformers, but they will all share that same, fixed active bond distance. As a result, for TSs that can adopt multiple active bond lengths, fixing the bond distance between forming or breaking atoms might exclude potential TS conformers with different active bond lengths, some of which could turn out to be the most stable TS. This issue becomes especially prominent for large, flexible

molecules. Because large systems possess more internal degrees of freedom, their TSs can exhibit high structural flexibility and may adopt different active bond lengths [34]. A clear example of this appears in the theoretical work by Medvedev and coworkers on the SpnF-catalyzed formal Diels–Alder cycloaddition [35, 36]. By systematically varying the fixed distance between the carbon atoms involved in ring formation, they generated multiple TSs with distinct active bond distances. In other words, they discovered additional TS structures that would have been overlooked if they had used a single, active bond constrained method. TS sampling approaches starting from multiple reactant/product structures provide a complementary strategy for exploring TS structures beyond a single prelocated TS geometry. Methods such as ConfoTS and Yet Another Reaction Program (YARP) perform double-ended TS searches using conformational ensembles of reactant and product structures and can thereby sample TS conformers with various active bond lengths [37], although they were not specifically developed for this purpose. Nevertheless, TS conformers that are stabilized only at the TS, such as those involving local charge separation or partial bond formation/cleavage, may be difficult to identify from reactant/product conformational ensembles alone. Building on these concepts, it would be valuable to further integrate the exploration of TS structures with different active bond lengths with the subsequent conformational sampling of the TS structures identified in the initial search. Such an integrated workflow would allow

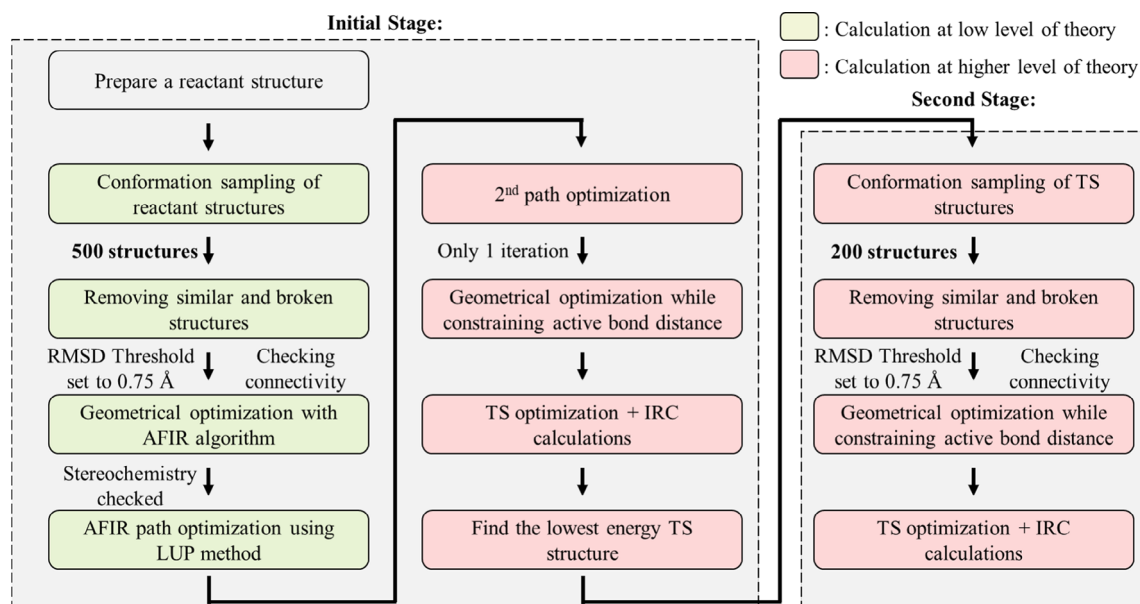


FIGURE 2 | Computational protocol of TStrail to generate TS conformers from a reactant structure.

variations in active bond lengths and conformational changes in non-reactive regions to be considered together in a systematic manner.

Here, we introduce TStrail, a comprehensive tool that integrates the exploration of TS structures with different active bond lengths with the subsequent conformational sampling of the TS structures identified in the initial search. TStrail first samples multiple conformations of reactant structures and then uses the AFIR function to generate reaction pathways, enabling the identification of TS structures with various active bond lengths (Figure 1B). The identified TS structures are then subjected to additional conformational sampling to explore conformational variations in non-reactive regions. Since TStrail does not require an initial TS structure, it would provide a broad range of TS structures with multiple active bond distances without any bias. This program also includes additional useful tools, such as automated stereocenter determination and TS convergence analysis, that ensure that the sampled TS conformers are actually the target structures. We have applied this method to various systems, from simple aldol reactions with different substituents and charges to a medium-sized (~100 atoms) chiral Brønsted acid-catalyzed asymmetric Mannich-type reaction to even larger (> 200 atoms) chiral Brønsted acid-catalyzed enantioselective intramolecular hydroamination of alkene. The generated TS conformers were further analyzed by studying the noncovalent interactions and the origins of enantioselectivity.

2 | Method

The procedure of TStrail can be broadly divided into two parts: (1) conformation sampling of reactant structures and generation of the corresponding reaction paths followed by optimization of the transition states, and (2) conformation sampling of the lowest-energy TS structure found in the previous step and further optimization to the TS structures (Figure 2). The initial stage is designed to identify TS structures with various active

bond lengths starting from reactant structures, whereas the second stage is intended to explore conformational variations of the individual TS structures identified in the initial stage.

TStrail requires the XYZ coordinates of the reactant structure as an input prepared in advance. Conformation sampling is performed on a reactant structure using the single-component AFIR (SC-AFIR) program implemented in the GRRM program [18, 38]. In the SC-AFIR program, fragment pairs are automatically defined within a given system and are pushed or pulled by the artificial force to induce possible reactions along the AFIR path. Subsequently, all local minima identified along the AFIR path are optimized to actual local minima on the PES. As the artificial-force-induced deformation frequently involves conformational rearrangements, SC-AFIR systematically samples a wide range of conformers accessible from the input structure. Thus, conformational rearrangements can be generated internally during reaction path exploration. The addition of the “NoBondRearrange” option further restricts the SC-AFIR procedure to structures with the same bond-connectivity matrix as that of the input structure, thereby focusing the search on conformational rearrangements within the same bonding pattern. Here, the distances between the reacting species are kept separated while the conformations are sampled, as detailed options used for this conformation sampling are described in the Supporting Information. In this study, conformation sampling of reactant structures was done until 500 conformers were searched, but this number can be set by users.

After conformation sampling of reactant structures, the program filters out any similar or broken structures that might be included in the searched conformers. Initially, the similar structures were removed by calculating the root-mean-square deviation (RMSD). The structures were first aligned pairwise using the Kabsch algorithm [39], and an RMSD matrix was built, where each cell (i, j) represents the RMSD between conformer i and conformer j . Agglomerative hierarchical clustering is then performed using this RMSD matrix with average

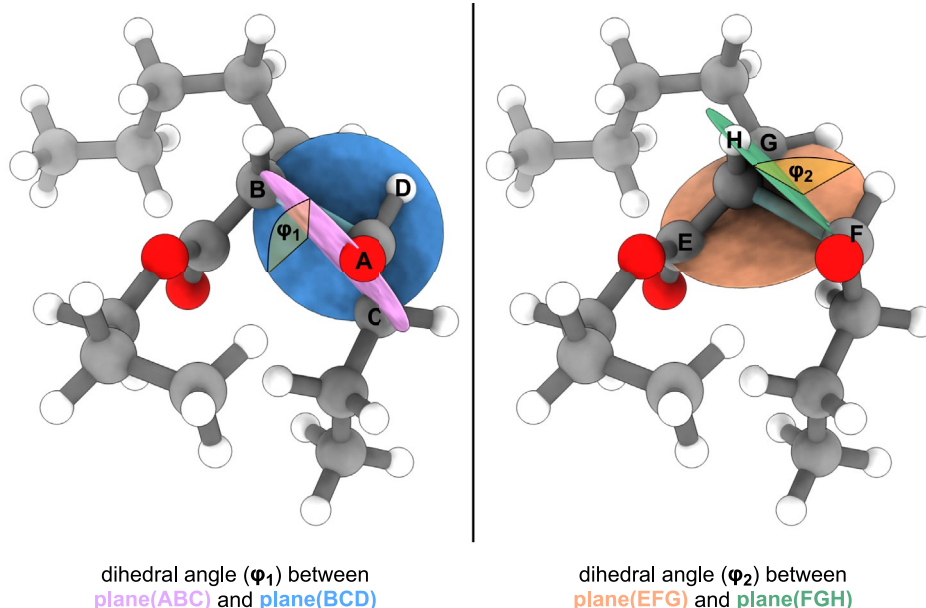


FIGURE 3 | Determination of product stereochemistry using dihedral angles. For products containing two stereocenters, two dihedral angles are calculated. In this example, dihedral angle φ_1 is defined between the plane formed by atoms A-B-C and the plane formed by atoms B-C-D. Similarly, dihedral angle φ_2 is defined between the plane formed by atoms E-F-G and the plane formed by atoms F-G-H. The calculated dihedral angles are then used to determine the stereochemistry; for instance, $-180^\circ < \varphi_1 < 0^\circ$ and $0^\circ < \varphi_2 < 180^\circ$ corresponds to the (1S, 2R) configuration.

linkage, in which the distance between two clusters is defined as the average of the pairwise RMSD values between conformers belonging to the two clusters (Figure S1). The resulting hierarchical clustering tree is cut at a user-defined RMSD threshold to determine the final conformer clusters. In this study, the RMSD threshold was set to 0.75 Å, and the lowest-energy conformer in each resulting cluster was selected as the representative conformer for the subsequent AFIR pathway search. This threshold can be easily modified by users, and further details of the clustering procedure are provided in the [Supporting Information](#). TStrail can automatically perform this clustering and generate input files for the next step. Secondly, the program also removes any broken structures that could be generated from the conformation sampling. This is done by setting the initial structure used for the reactant sampling as a reference and comparing the bond connectivity with the other reactant conformers. All the structures that do not share the same bond connectivity with the reference structure are automatically eliminated, giving only appropriate reactant conformers that proceed to the next stage.

After the reactant conformers have been searched, TStrail calculates approximate TS structures using the AFIR algorithm. The atoms that are involved in bond formation are specified by users. The AFIR algorithm then applies a constant force on these active bonds during a DFT geometry optimization, drawing together the atoms that form new bonds [40]. This process generates a reaction pathway, defined as the AFIR pathway, which connects the reactant structure, through the transition state, to the product structure. To reduce the computational cost, the AFIR search itself was terminated after 30 geometry optimization iterations under the artificial force, rather than performing a fully converged AFIR optimization. It should be noted that, because TStrail conducts conformational sampling of the reactant structures at relatively long interatomic distances, it

can explore pathways leading to all stereoisomers from a single reactant. Therefore, this program enables sampling of multiple stereoisomeric pathways, which would be highly valuable in the computational studies of stereoselective transformations. To minimize the calculation cost for undesired stereoisomers, pathways that result in undesired products are removed. Therefore, after generating a pathway, the stereochemistry of the last structure in that pathway is checked. This is accomplished by calculating the dihedral angles between the two reactant species, as depicted in Figure 3. For reactions in which only a single stereocenter is formed, calculating one dihedral angle is sufficient.

After confirming that each AFIR pathway leads to the desired product, the pathways are optimized using the locally updated plane (LUP) method implemented in the GRRM program. In the LUP method, a reaction path is optimized by evenly distributing path points along the path and relaxing them toward lower energy regions on hyperplanes perpendicular to the path tangent. Although one could simply take the highest-energy structure from the unoptimized AFIR pathway as a TS-like structure and directly perform a TS optimization, using the highest-energy point of the optimized AFIR pathway typically converges to the TS structure more quickly and accurately. As noted earlier, the conformational sampling of the reactant structures, the geometrical optimization using the AFIR algorithm, and the path optimization are all carried out at the user-specified computationally inexpensive level of theory, in this case, GFN2-xTB.

For the final part of the initial stage of TStrail, the AFIR pathways computed at a lower level of theory are re-optimized at a higher level of theory, followed by TS optimization. First, the AFIR pathways are used as input for another LUP calculation. To reduce computational costs, only a single iteration of path optimization is performed for each pathway. However, the number of iterations can be easily modified by changing the

“MaxLUPITR” value in the option file. Next, the local maxima of each AFIR pathway are selected, and geometry optimizations are conducted while keeping the active bond distances fixed. This step helps lower the cost of the subsequent TS optimization. Finally, the resulting TS-like structures are optimized to obtain the actual TS at the same level of theory as the constrained geometrical optimization. For all TS calculations, intrinsic reaction coordinate (IRC) calculations are also performed to verify that each TS indeed connects the correct reactant and product. Finally, the energy of the calculated TS conformers is checked, and the lowest TS structure is used for subsequent TS sampling. Additional details and protocols for TS optimizations are described in the [Supporting Information](#).

The key for enabling TStrail to identify TS structures with different active bond lengths in the initial stage of TStrail is the conformation sampling of reactant structures. The calculation of reactant conformer ensemble could generate diverse reactant structures with different distances and orientations between the reacting atoms, as well as different surrounding conformations. These differences produce multiple AFIR pathways leading to the corresponding product. Through subsequent pathway relaxation and TS optimization calculations, TStrail can identify TS structures located at different positions along the reaction coordinate and, consequently, with different active bond lengths.

In the second stage of TStrail, the lowest energy TS from the initial stage is used as an input for subsequent TS conformation sampling. The process is very similar to the conventional TS sampling, in which the active bond length is fixed while conformation sampling is performed using the SC-AFIR program. Sampling was done until 200 unique TS-like structures were obtained, where duplicates were removed by applying an RMSD threshold of 0.75 Å. Each TS-like structure was then subjected to constrained optimization, followed by TS optimization and an IRC calculation to produce the final TS structures. In the end, TStrail compiles the TS energies, active bond lengths, and imaginary frequencies in a single text file for further analysis.

3 | Applications

3.1 | Application 1: Aldol Reactions

Initially, we applied TStrail to three model aldol reactions, each featuring different charge states and substituent groups, as illustrated in Figure 4 and Figure S2. Although four stereoisomers can form after C—C bond formation, we only focused on the TS structures that lead to (1*S*, 2*R*) product.

For the reaction between an aldehyde and an enol, the initial stage of TStrail located 145 unique TS conformers as shown in Figure 4B. Their C—C distances ranged from 2.07 to 2.16 Å, and the highest-energy TS was 5.3 kcal/mol higher than the lowest TS conformer. Because the aldehyde and enol can adopt a stable six-membered ring TS structure, as commonly described as Zimmerman-Traxler model [41], the distribution of active bond lengths is very narrow (Figure 4G). Comparison of the lowest- and highest-energy TS conformers shows that the orientation of the long alkyl chain of the enol differs substantially (Figure 4E).

The resulting distortion of the enol, together with steric repulsion between the enol and aldehyde fragments, may contribute to the relative energy difference (Table S1).

In stark contrast, when an enolate anion is employed as a reactant, the initial search produced 157 unique TS conformers whose C—C distances ranged from 1.96 to 2.42 Å with the most stable TS conformer having an active bond length of 2.22 Å (Figure 4C). Removal of a hydrogen atom from the enol altered the electronic environment, and TStrail was able to sample TS conformations for reactions where the conformers do not share the same active bond distance. Regarding the origin of the relative energy difference (Figure 4F), conformational variations of the long alkyl chain in the enolate anion may influence the TS energy, as observed for the reaction between the aldehyde and enol (Table S2).

Another type of aldol reaction was tested using enolate anion from ester in order to further modify the chemical environment (Figure 4D). The initial round of TStrail generated 140 unique TS conformers. By using ester-derived enolate anion, widening of the TS active bond lengths was observed, as the shortest TS conformer had a C—C length of 2.06 Å while the longest TS conformer had an active bond length of 2.89 Å. TStrail was able to successfully capture this wide range of active bond lengths since it does not rely on an initial TS structure that might bias conformer sampling.

From the reaction with a distinct active bond distance to reactions with multiple active bond distances, TStrail successfully identified TS conformers by sampling the reaction pathways (Figure 5). We believe that TStrail is therefore well suited for sampling TS conformers in reactions where the active bond distance may deviate from a pre-defined distance or vary across different TS structures. By avoiding reliance on a single pre-determined TS geometry, TStrail enables broader and less biased exploration of TS conformational space, making it a promising approach for studying reactions with complex or flexible TS landscapes.

3.2 | Application 2: BINOL-Catalyzed Mannich-Type Reaction

Next, TStrail was applied to chiral binaphthol (BINOL)-derived phosphoric acid catalyzed Mannich-type reaction reported by Yamanaka and coworkers in 2007 [42]. In this reaction, (*R*)-BINOL phosphoric acid was used as a catalyst for the reaction between an aldimine and a silyl ketene acetal to produce a highly enantioselective adduct in toluene at −78°C (Figure 6A). We chose this reaction as a demonstration for TStrail since the noncovalent interactions between the catalyst and substrate could generate a wide variety of conformers, and sampling of these various catalyst-substrate complexes is likely the key to calculating the enantioselectivity at high accuracy.

In this study, we applied TStrail to the carbon-carbon bond forming step, since this step is considered as an enantiodetermining step from the previous study [43]. Conformational searches were performed on the reactant complexes leading to

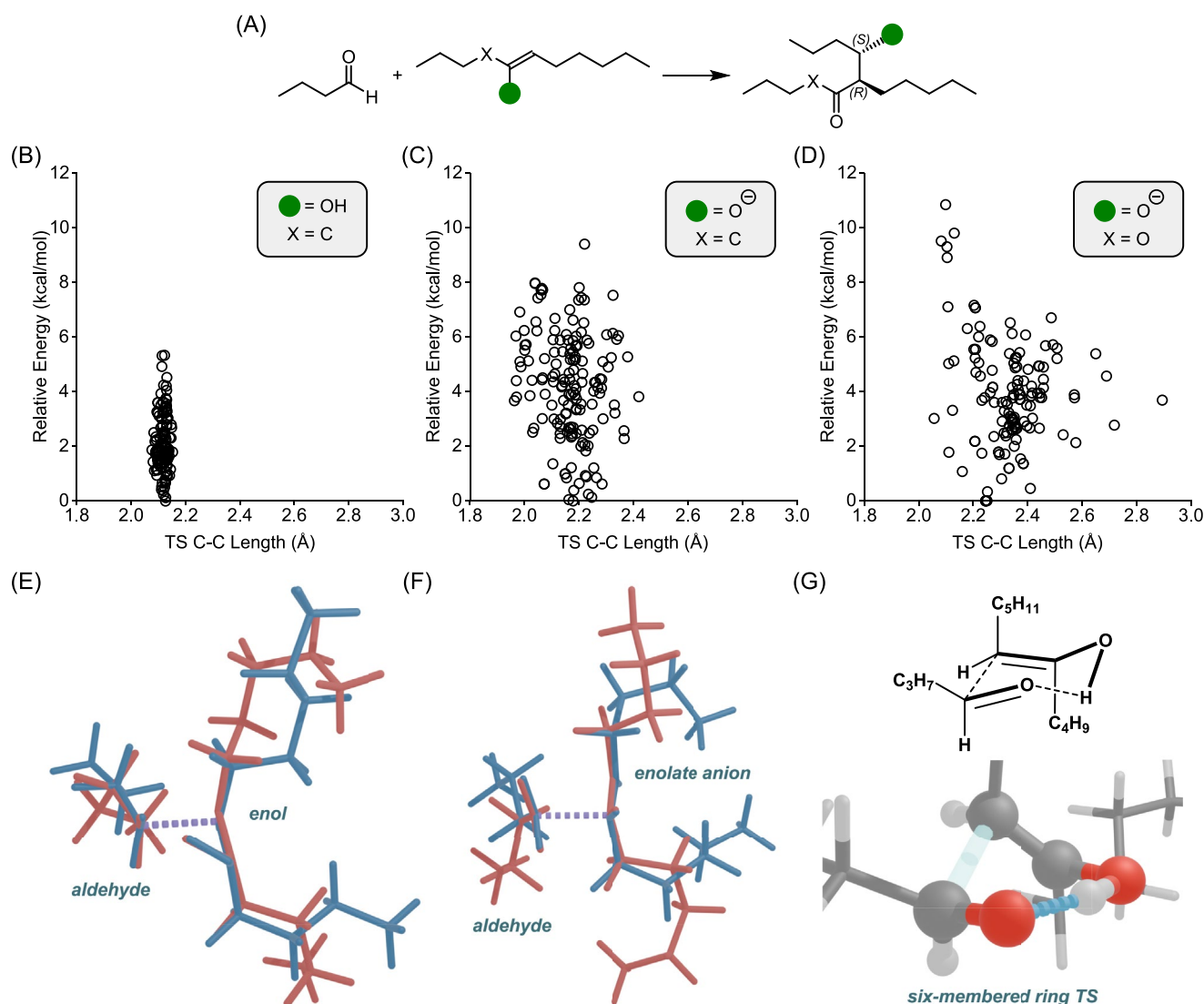


FIGURE 4 | (A) TStrail applied to Aldol reactions that will yield (1*S*,2*R*)-products. (B–D) The initial stage of TStrail results for three aldol reactions are represented. Relative energies of TS conformers are plotted against the C–C bond distances. Each black circle denotes a single conformer. (E) TS conformers corresponding to the aldehyde-enol reaction in panel (B). The most stable TS conformer is shown as blue sticks, whereas the least stable TS conformer is shown as red sticks (+5.3 kcal/mol relative to the most stable TS). (F) TS conformers corresponding to the aldehyde-enolate anion reaction in panel (C). The most stable TS conformer is shown as blue sticks, whereas the least stable TS conformer is shown as red sticks (+9.4 kcal/mol relative to the most stable TS). (G) Visualization of the six-membered ring TS structure.

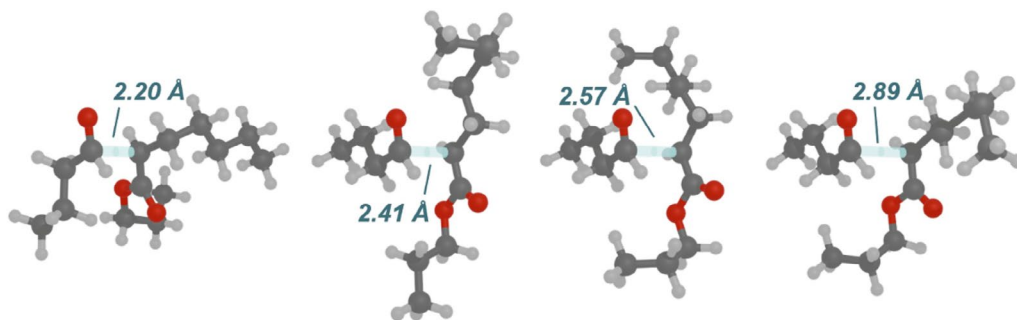


FIGURE 5 | Representative TS conformers from Figure 4D. Four representative TS conformers with different active bond lengths, ranging from 2.20 to 2.89 Å. The light-blue bonds between the bond-forming carbon atoms indicate the active bond lengths.

both the (*S*)- and (*R*)-products. In the initial stage of TStrail, 213 (*S*)-TS conformers were calculated while 208 (*R*)-TS conformers were calculated (Figure 6B). Here, we see that TStrail was

able to sample TS conformers with various active bond lengths: the TS C–C bond distance distributed from 2.07 Å to 2.55 Å for (*S*)-TSS while calculated (*R*)-TS conformers had 2.03 Å to 2.30 Å.

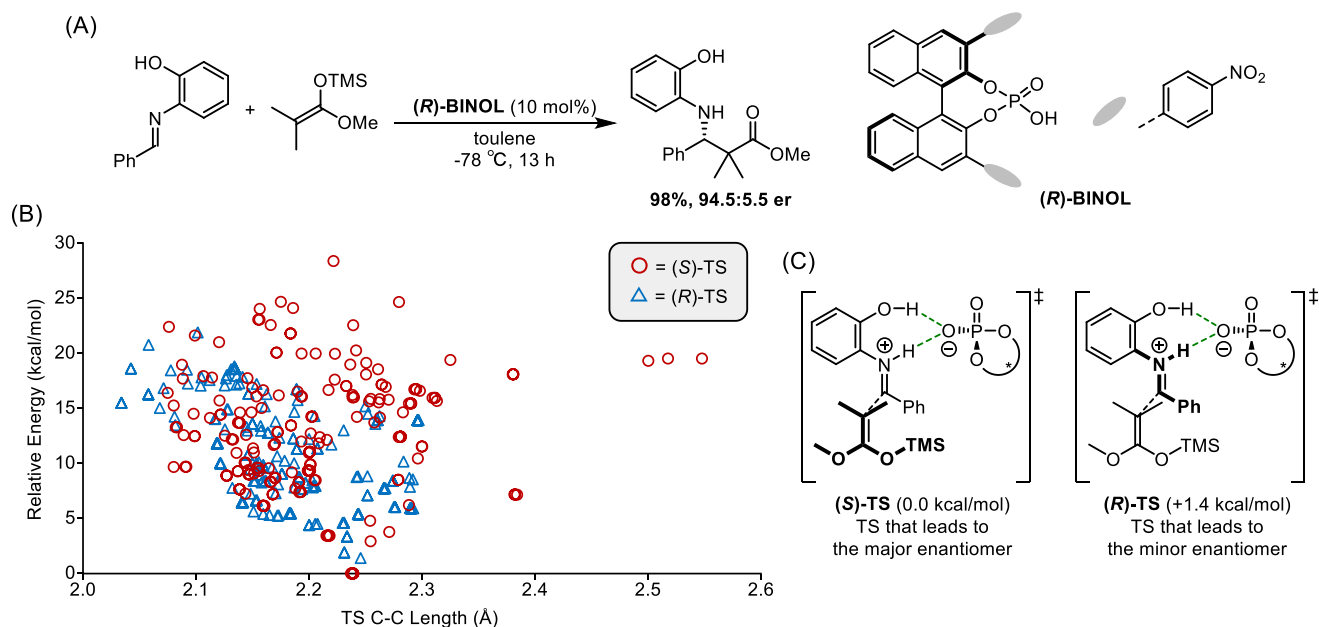


FIGURE 6 | (A) BINOL-catalyzed Mannich-type reaction. The experimental data were obtained from Yamanaka et al. (2007). (B) The result from the initial round of TStrail, where the red circle represents (S)-TSs that will lead to the major enantiomer while the blue triangle represents (R)-TSs that will lead to the minor enantiomer. (C) TS conformer of most stable TS from the initial stage of TStrail with relative electronic energy.

TStrail was also able to capture TSs with substantially different energies, with many TS conformers lying 5–20 kcal/mol higher than the lowest-energy TS conformer and the highest-energy TS being 28.3 kcal/mol higher. These energy differences are larger than those observed in the previous application case, further highlighting the value of TStrail for capturing such diverse TS geometries. When comparing the lowest-energy TS conformers of the (S)- and (R)-TSs, the energy difference was approximately 1.4 kcal/mol (Figure 6C).

To evaluate the enantioselectivity, TS conformers obtained from the second stage of TStrail were used, in which 103 (S)-TS conformers were calculated while 91 (R)-TS conformers were obtained (Figure S3). The free energies were calculated for all obtained transition states. Then, the enantioselectivity was calculated based on the Boltzmann distribution (Figure S4), which gave $\Delta\Delta G^\ddagger = 0.82$ kcal/mol (89.2:10.8 e.r.), which is in good agreement with the experimental values ($\Delta\Delta G^\ddagger = 1.10$ kcal/mol, 94.5:5.5 e.r.).

We then turned our attention to the calculated TS complexes of the present reaction. In the previous study by Akiyama and coworkers [42], they proposed two possible TS complexes (Figure 7A): (i) a monocoordination complex in which one phosphate oxygen forms hydrogen bonds with the imine N–H and O–H groups, generating a seven-membered cyclic TS, and (ii) a dicoordination complex in which both phosphate oxygens form hydrogen bonds with the aldimine, giving a nine-membered cyclic TS. In this work, the initial stage of TStrail was able to capture both complexes for (S)-TSs and (R)-TSs, as well as a 12-membered ring TS complex for (S)-TS where two phosphoryl oxygens form hydrogen bonds to the imine O–H and to a phenyl C–H of the aldimine (Figure 7B,C). By using TStrail, we found that the monocoordination complex is more stable than the dicoordination complex for both (S)- and (R)-TS conformers. Notably, the 7-membered cyclic TS and the 9-membered cyclic TS show

opposite stereochemical preferences, indicating that incomplete TS conformer sampling could lead to an incorrect prediction of the stereochemical outcome. This result highlights the necessity of exhaustive TS conformer sampling to locate low-energy TSs and correctly evaluate stereochemical preferences.

3.3 | Application 3: IDPi-Catalyzed *Anti*-Selective Hydroamination of Alkenes

Finally, we have applied TStrail to imidodiphosphorimidates (IDPi) catalyzed intramolecular hydroamination reaction reported by Guria and coworkers (Figure 8A) [44]. Here, the IDPi catalyst was used as a Brønsted acid to catalyze the intramolecular hydroamination of 1,1-disubstituted alkenes with high enantioselectivities. The structures reported in the previous computational study were used as starting points to recalculate the TSs at a higher level of theory (see the Supporting Information for the full analysis). The initial calculations indicated that the stepwise pathway is energetically more favorable than the previously reported fully concerted pathway (Figure S5). In a stepwise pathway, protonation from the IDPi catalyst activates the alkene and generates a tertiary carbocation intermediate that is stabilized inside the microenvironment offered by a chiral counteranion. This intermediate then undergoes the enantio-determining cyclization step by forming a C–N bond in an intramolecular manner. We envisioned that TStrail is well suited for this reaction because TS structures involving ionic species, which typically exhibit relatively flat PES and small imaginary frequencies, are rather challenging to sample its TS conformations using conventional methods due to their variable active bond lengths, whereas TStrail can handle them effectively.

As described in the Computational Details section, our in-house GRRMOnioM program was employed for the higher-level calculations in TStrail (Figure S6). In the initial stage of TStrail, 56

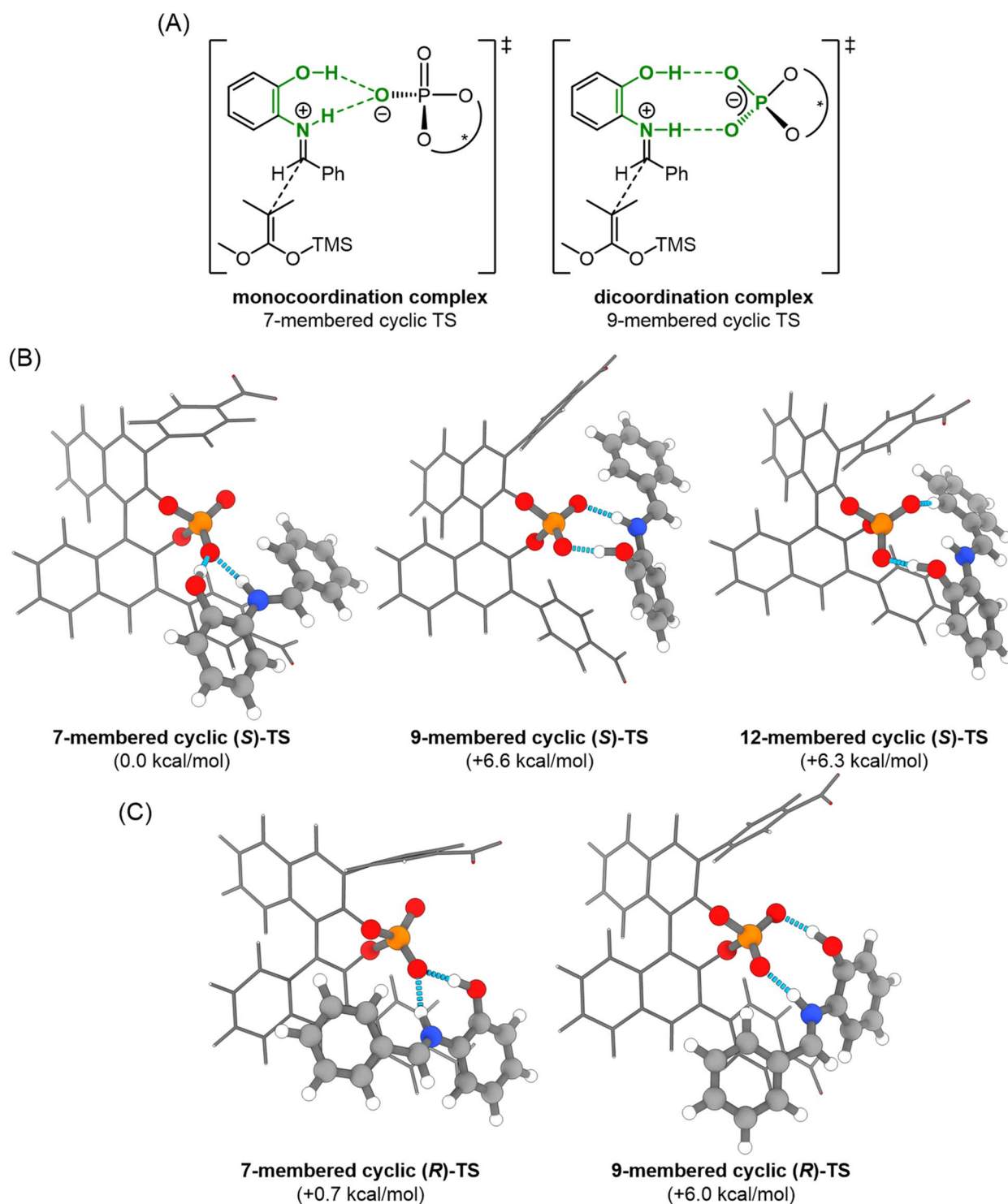


FIGURE 7 | (A) Proposed TS complexes from previous research. The atoms highlighted in green constitute the cyclic TS. (B) Calculated (S)-TS complexes from the second round of TStrail. (C) Calculated (R)-TS complexes from second round of TStrail. All energies were calculated at the CPCM(Toluene)- ω B97X-V/def2-TZVPP// ω B97X-D/def2-SV(P) level of theory. The structure of silyl ketene acetal is removed for visual purposes.

TS conformers leading to the (*R*)-product and 49 TS conformers leading to the (*S*)-product were obtained, while sampling 500 reactant conformers. The C—N bond distances of the TS conformers and their relative energies are summarized in Figure 8B. The smaller number of TS conformers obtained at this stage compared with the previous cases reflects the inherent difficulty of sampling reactive ionic complexes. The carbocationic intermediate is highly reactive and can transiently form additional

bonds with the catalyst or undergo intramolecular reactions during conformational exploration. To ensure that only chemically meaningful reaction pathways were analyzed, such structures were systematically removed during the data-processing step. After eliminating redundant and chemically irrelevant structures, 131 AFIR pathways leading to the (*R*)-product and 123 AFIR pathways leading to the (*S*)-product were retained for further analysis. From this, some TS optimizations failed to

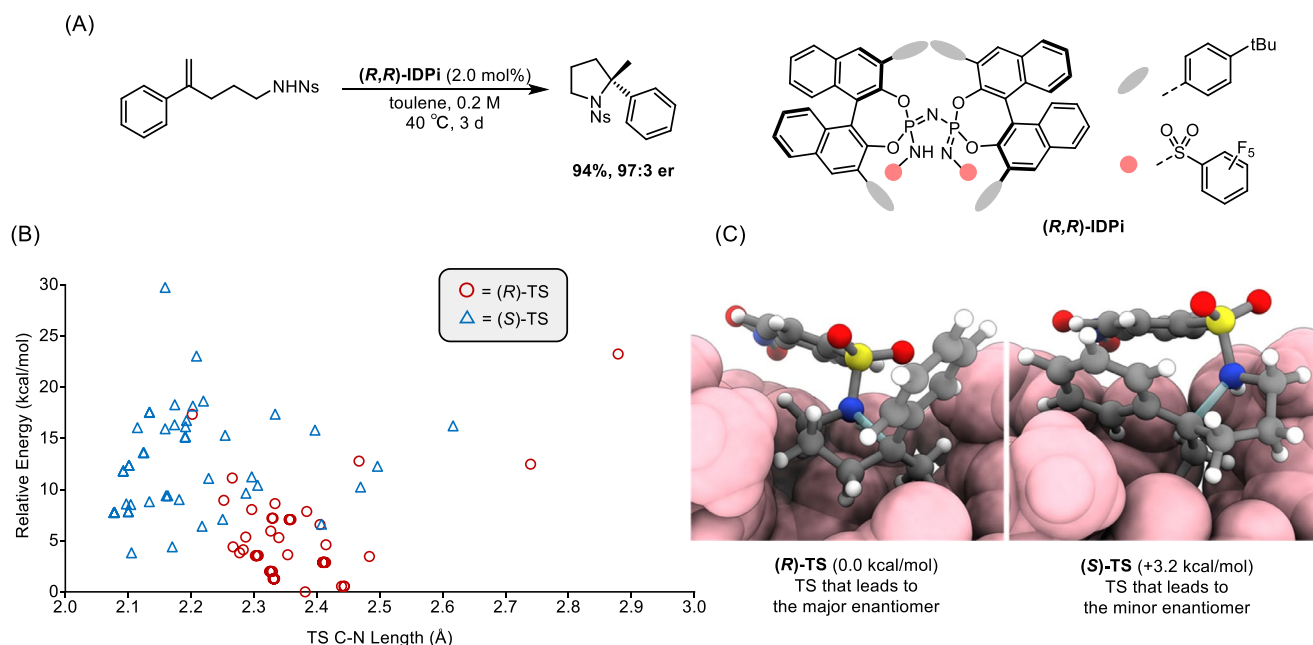


FIGURE 8 | (A) IDPi-catalyzed intramolecular hydroamination reaction. The experimental data were obtained from Guria et al. (2024). (B) The result from the initial round of TStrail, where the red circle represents (R)-TSs that will lead to the major enantiomer while the blue triangle represents (S)-TSs that will lead to the minor enantiomer. (C) TS conformer of most stable TS from the initial stage of TStrail with relative electronic energy.

converge due to several factors. For example, the reactant contains rotatable groups, such as the nosyl and phenyl groups, and unfavorable orientations of these groups may decrease the TS convergence rate. Additional factors include non-convergence of the TS optimization within the predefined threshold, failure of the IRC calculation to connect to the desired reactant and product structures, and imaginary frequencies of the optimized TS structures being below the predefined minimum threshold. The energy difference between the lowest-energy TS conformers of the two TSs was 3.2 kcal/mol (Figure 8C).

The lowest energy TS conformer from each enantiomer was then used for the second stage of TStrail. After sampling 200 TS-like structures for each enantiomer, 15 (R)-TS conformers and 18 (S)-TS conformers were obtained (Figure S7). TS conformers with energies within 5 kcal/mol of the lowest energy TS conformer were subsequently re-optimized at the r^2 SCAN-3c level of theory, yielding 5 (R)-TS conformers and 8 (S)-TS conformers. By using the Boltzmann distribution to calculate the enantioselectivity (Figure S8), it was found that the calculated energy difference between the (R)-TS and (S)-TS was $\Delta\Delta G^\ddagger = 2.15$ kcal/mol (97:3 e.r.), which is in good agreement with the experimental values ($\Delta\Delta G^\ddagger = 2.16$ kcal/mol, 97:3 e.r.). Additional three-dimensional views of the TS conformers are represented in Figures S9 and S10.

To elucidate the origin of enantioselectivity, distortion-interaction analysis was performed on the lowest energy (R)-TS and (S)-TS structure from each TS conformer [45]. This analysis suggested that the energy difference between the two enantiodetermining TSs mainly arises from differences in the catalyst-substrate interactions (Table S3). To further investigate this difference in the weak interactions, the independent gradient model based on the Hirshfeld partition (IGMH) method

was applied using the Multiwfn program [46–48]. Each TS was divided into two fragments: the cationic substrate and the anionic IDPi catalyst, and the weak interactions between these fragments were analyzed. The results suggest that the difference between the two enantiodetermining TSs primarily originates from the interaction between the NH-proton of the substrate and the sulfonyl group of the catalyst (Figure 9). In the (R)-TS, the NH-proton is oriented toward the inside of the anionic catalytic pocket, forming a strong NH/O interaction. In contrast, the NH proton is directed away from the catalytic pocket for the (S)-TS, resulting in weaker interactions and thereby leading to the observed energy difference (Table S4).

Free-energy diagrams for the three aldol reactions, the BINOL-catalyzed Mannich-type reaction, and the IDPi-catalyzed intramolecular hydroamination reaction are provided in Figures S11, S12, and S13, respectively.

4 | Summary and Outlook

Here, we report a comprehensive TS conformational sampling tool designed to systematically capture TS conformers that differ not only in their active bond lengths but also in the conformations of non-reactive regions. TStrail first performs conformational sampling of the reactant structure to identify TS conformers with different active bond distances. It then conducts conformational sampling of the TS structures obtained in the initial stage to explore additional conformational variations of individual TS structures. The developed program was first applied to the aldol reaction to verify its ability to sample TS conformers with different active bond lengths, and was subsequently applied to the Mannich-type reaction and the intramolecular hydroamination reaction to demonstrate its broader

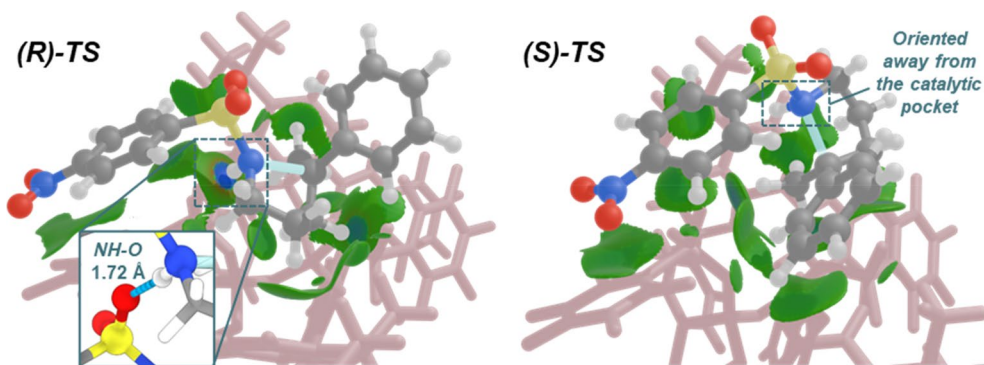


FIGURE 9 | Visualization of noncovalent interactions between a cationic substrate and an anionic IDPi catalyst for the lowest energy (R)-TS (left) and (S)-TS (right) from each conformer. A green isosurface represents dispersion interactions.

applicability. Without constraining the active bond length, TStrail successfully identified low-energy TS conformers that were consistent with experimental outcomes. TStrail is in principle applicable to any reaction step in which the key bond-forming or bond-breaking event is specified in advance.

We believe that one of the key areas of application for TStrail is TS conformational sampling of large, flexible, ionic, and complex TS structures. Such TS conformers often involve conformational variations not only in the active bond distances but also in the non-reactive regions, and their relatively flat PESs make exhaustive exploration challenging using conventional strategies based on fixing the active bond length. By combining TStrail with faster computational approaches, such as machine-learning potentials, it should be possible to efficiently perform TS conformational sampling for such challenging large and complex chemical transformations.

5 | Computational Details

All calculations in this work, except for single-point calculations, were performed using the developer version of the Global Reaction Route Mapping (GRRM) software [49, 50]. Since TStrail uses the GRRM program as an interface, users can readily adopt any suitable level of theory with various DFT packages, such as Gaussian, ORCA, TURBOMOLE, and others. The program can also be suited to some of the newly developed machine learning potentials, such as the atoms-in-molecules neural network potential (AIMNet2) and Universal Models for Atoms (UMA) [51, 52]. The input files for calculations using various software can be found on GitHub.

In this study, the GFN2-xTB level of theory implemented in the ORCA 4.2.0 program was used for the conformation sampling and the initial path optimization [53, 54]. Geometry optimizations and TS optimizations for the aldol reactions and the BINOL-catalyzed Mannich-type reaction were carried out at the ω B97X-D/def2-SV(P) level of theory using the Gaussian 16 package [55–58]. Due to the high computational cost of Brønsted acid-catalyzed intramolecular hydroamination reaction, an in-house ONIOM program was initially used for the calculation of TS structures [59–62], where ω B97X-D/def2-SV(P) level of theory was used for the high-layer and GFN2-xTB level of theory was used for the low-layer with xtb version

6.4.1 [63, 64]. Then, only TS conformers within 5 kcal/mol of the lowest energy TS conformer, which are expected to contribute to the Boltzmann-weighted enantioselectivity, were re-optimized at the r^2 SCAN-3c level of theory implemented in the ORCA 5.0.3 program [65, 66]. The single-point energies were computed using the ω B97X-V functional with the def2-TZVPP basis set [67, 68], including solvation effects described by the CPCM(Toluene) model as implemented in ORCA [69]. Free energies were evaluated at 195.15 K and 1 atm for the Mannich-type reaction and at 313.15 K and 1 atm for the intramolecular hydroamination reaction. Transition state structures were visualized using UCSF Chimera and ChimeraX version 1.7 followed by rendering with Blender version 4.2 [70–72].

Acknowledgments

This work was supported by the Japan Science and Technology Agency (JST) SPRING JPMJSP2119, the Institute for Chemical Reaction Design and Discovery (ICReDD), which was established by the World Premier International Research Initiative (WPI), MEXT, Japan, and the List Sustainable Digital Transformation Catalyst Collaboration Research Platform offered by Hokkaido University. Part of the computation was performed using the Research Center for Computational Science, Okazaki, Japan (Projects 24-IMS-C114 and 25-IMS-C115) as well as the supercomputer of ACCMS, Kyoto University.

Funding

This work was supported by Japan Science and Technology Agency, JPMJSP2119.

Data Availability Statement

The code for TStrail is available at (<https://doi.org/10.5281/zenodo.21252469>). The input and output files for all TS calculations are available in the ioChem-BD repository at (<https://iochem-bd.bsc.es/browse/review-collection/100/488944/39facfa781c115de14107f88>) [73].

References

1. H. B. Schlegel, “Exploring Potential Energy Surfaces,” *Pure and Applied Chemistry* 97 (2025): 1047–1063.
2. S. Maeda, Y. Harabuchi, Y. Ono, T. Taketsugu, and K. Morokuma, “Intrinsic Reaction Coordinate: Calculation, Bifurcation, and Automated Search,” *International Journal of Quantum Chemistry* 115 (2015): 258–269.

3. A. L. Dewyer, A. J. Argüelles, and P. M. Zimmerman, "Methods for Exploring Reaction Space in Molecular Systems," *WIREs Computational Molecular Science* 8 (2018): e1354.
4. M. Foscato and V. R. Jensen, "Automated In Silico Design of Homogeneous Catalysts," *ACS Catalysis* 10 (2020): 2354–2377.
5. G. N. Simm, A. C. Vaucher, and M. Reiher, "Exploration of Reaction Pathways and Chemical Transformation Networks," *Journal of Physical Chemistry A* 123 (2019): 385–399.
6. J. N. Harvey, F. Himo, F. Maseras, and L. Perrin, "Scope and Challenge of Computational Methods for Studying Mechanism and Reactivity in Homogeneous Catalysis," *ACS Catalysis* 9 (2019): 6803–6813.
7. F. Maseras and K. Morokuma, "IMOMM: A New Integrated Ab Initio + Molecular Mechanics Geometry Optimization Scheme of Equilibrium Structures and Transition States," *Journal of Computational Chemistry* 16 (1995): 1170–1179.
8. D. J. Wales and T. V. Bogdan, "Potential Energy and Free Energy Landscapes," *Journal of Physical Chemistry B* 110 (2006): 20765–20776.
9. H. B. Schlegel, "Exploring Potential Energy Surfaces for Chemical Reactions: An Overview of Some Practical Methods," *Journal of Computational Chemistry* 24 (2003): 1514–1527.
10. P. M. Zimmerman, "Single-Ended Transition State Finding With the Growing String Method," *Journal of Computational Chemistry* 36 (2015): 601–611.
11. E. Martínez-Núñez, "An Automated Method to Find Transition States Using Chemical Dynamics Simulations," *Journal of Computational Chemistry* 36 (2015): 222–234.
12. L.-P. Wang, A. Titov, R. McGibbon, F. Liu, V. S. Pande, and T. J. Martínez, "Discovering Chemistry With an *ab Initio* Nanoreactor," *Nature Chemistry* 6 (2014): 1044–1048.
13. C. Lavigne, G. Gomes, R. Pollice, and A. Aspuru-Guzik, "Guided Discovery of Chemical Reaction Pathways With Imposed Activation," *Chemical Science* 13 (2022): 13857–13871.
14. G. N. Simm and M. Reiher, "Context-Driven Exploration of Complex Chemical Reaction Networks," *Journal of Chemical Theory and Computation* 13 (2017): 6108–6119.
15. G. Li, Z. Li, L. Gao, S. Chen, G. Wang, and S. Li, "Combined Molecular Dynamics and Coordinate Driving Method for Automatically Searching Complicated Reaction Pathways," *Physical Chemistry Chemical Physics* 25 (2023): 23696–23707.
16. I. Berente and G. Náray-Szabó, "Multicoordinate Driven Method for Approximating Enzymatic Reaction Paths: Automatic Definition of the Reaction Coordinate Using a Subset of Chemical Coordinates," *Journal of Physical Chemistry A* 110 (2006): 772–778.
17. K. Lee, J. H. Kim, and W. Y. Kim, "pyMCD: Python Package for Searching Transition States via the Multicoordinate Driven Method," *Computer Physics Communications* 291 (2023): 108831.
18. S. Maeda, Y. Harabuchi, M. Takagi, T. Taketsugu, and K. Morokuma, "Artificial Force Induced Reaction (AFIR) Method for Exploring Quantum Chemical Potential Energy Surfaces," *Chemical Record* 16 (2016): 2232–2248.
19. E. Weinan, W. Ren, and E. Vanden-Eijnden, "String Method for the Study of Rare Events," *Physical Review B* 66 (2002): 052301.
20. G. Henkelman and H. Jónsson, "Improved Tangent Estimate in the Nudged Elastic Band Method for Finding Minimum Energy Paths and Saddle Points," *Journal of Chemical Physics* 113 (2000): 9978–9985.
21. P. Zimmerman, "Reliable Transition State Searches Integrated With the Growing String Method," *Journal of Chemical Theory and Computation* 9 (2013): 3043–3050.
22. T. A. Young, J. J. Silcock, A. J. Sterling, and F. Duarte, "autodE: Automated Calculation of Reaction Energy Profiles— Application to Organic and Organometallic Reactions," *Angewandte Chemie International Edition* 60 (2021): 4266–4274.
23. S. Grimme, "Exploration of Chemical Compound, Conformer, and Reaction Space With Meta-Dynamics Simulations Based on Tight-Binding Quantum Chemical Calculations," *Journal of Chemical Theory and Computation* 15 (2019): 2847–2862.
24. Y. Fujino, K. Fukaya, and D. Urabe, "Automated Reactivity Prediction of Complex Molecules and Its Application to Total Synthesis of Ageliferin," *Journal of the American Chemical Society* 147 (2025): 34556–34563.
25. C. Peng and H. B. Schlegel, "Combining Synchronous Transit and Quasi-Newton Methods to Find Transition States," *Israel Journal of Chemistry* 33 (1993): 449–454.
26. A. Behn, P. M. Zimmerman, A. T. Bell, and M. Head-Gordon, "Efficient Exploration of Reaction Paths via a Freezing String Method," *Journal of Chemical Physics* 135 (2011): 224108.
27. E. Hansen, A. R. Rosales, B. Tutkowski, P.-O. Norrby, and O. Wiest, "Prediction of Stereochemistry Using Q2MM," *Accounts of Chemical Research* 49 (2016): 996–1005.
28. D. Balcells and F. Maseras, "Computational Approaches to Asymmetric Synthesis," *New Journal of Chemistry* 31 (2007): 333–343.
29. T. Kamachi and K. Yoshizawa, "Low-Mode Conformational Search Method With Semiempirical Quantum Mechanical Calculations: Application to Enantioselective Organocatalysis," *Journal of Chemical Information and Modeling* 56 (2016): 347–353.
30. P. Pracht, F. Bohle, and S. Grimme, "Automated Exploration of the Low-Energy Chemical Space With Fast Quantum Chemical Methods," *Physical Chemistry Chemical Physics* 22 (2020): 7169–7192.
31. Y. Guan, V. M. Ingman, B. J. Rooks, and S. E. Wheeler, "AARON: An Automated Reaction Optimizer for New Catalysts," *Journal of Chemical Theory and Computation* 14 (2018): 5249–5261.
32. H. Yang and M. W. Wong, "Automatic Conformational Search of Transition States for Catalytic Reactions Using Genetic Algorithm," *Journal of Physical Chemistry A* 123 (2019): 10303–10314.
33. S. P. Schmid, H. Seng, T. Kläy, and K. Jorner, "Rapid Generation of Transition-State Conformer Ensembles via Constrained Distance Geometry," *Journal of Chemical Information and Modeling* 66 (2026): 2777–2790.
34. M. Hatanaka, S. Maeda, and K. Morokuma, "Sampling of Transition States for Predicting Diastereoselectivity Using Automated Search Method—Aqueous Lanthanide-Catalyzed Mukaiyama Aldol Reaction," *Journal of Chemical Theory and Computation* 9 (2013): 2882–2886.
35. M. G. Medvedev, A. A. Zeifman, F. N. Novikov, et al., "Quantifying Possible Routes for SpnF-Catalyzed Formal Diels-Alder Cycloaddition," *Journal of the American Chemical Society* 139 (2017): 3942–3945.
36. M. G. Medvedev, M. V. Panova, G. G. Chilov, et al., "Exhaustive Conformational Search for Transition States: The Case of Catechol O-Methyltransferase Active Site," *Mendeleev Communications* 27 (2017): 224–227.
37. Q. Zhao and B. M. Savoie, "Simultaneously Improving Reaction Coverage and Computational Cost in Automated Reaction Prediction Tasks," *Nature Computational Science* 1 (2021): 479–490.
38. S. Maeda, T. Taketsugu, and K. Morokuma, "Exploring Transition State Structures for Intramolecular Pathways by the Artificial Force Induced Reaction Method," *Journal of Computational Chemistry* 35 (2014): 166–173.
39. W. Kabsch, "A Solution for the Best Rotation to Relate Two Sets of Vectors," *Acta Crystallographica, Section A: Crystal Physics, Diffraction, Theoretical and General Crystallography* 32 (1976): 922–923.
40. S. Maeda and K. Morokuma, "Finding Reaction Pathways of Type A + B → X: Toward Systematic Prediction of Reaction Mechanisms," *Journal of Chemical Theory and Computation* 7 (2011): 2335–2345.

41. H. E. Zimmerman and M. D. Traxler, "The Stereochemistry of the Ivanov and Reformatsky Reactions. I," *Journal of the American Chemical Society* 79 (1957): 1920–1923.
42. M. Yamanaka, J. Itoh, K. Fuchibe, and T. Akiyama, "Chiral Brønsted Acid Catalyzed Enantioselective Mannich-Type Reaction," *Journal of the American Chemical Society* 129 (2007): 6756–6764.
43. I. D. Gridnev, M. Kouchi, K. Sorimachi, and M. Terada, "On the Mechanism of Stereoselection in Direct Mannich Reaction Catalyzed by BINOL-Derived Phosphoric Acids," *Tetrahedron Letters* 48 (2007): 497–500.
44. S. Guria, A. N. Volkov, R. Khudaverdyan, et al., "Enantioselective, Brønsted Acid-Catalyzed Anti-Selective Hydroamination of Alkenes," *Journal of the American Chemical Society* 146 (2024): 17180–17188.
45. F. M. Bickelhaupt and K. N. Houk, "Analyzing Reaction Rates With the Distortion/Interaction-Activation Strain Model," *Angewandte Chemie International Edition* 56 (2017): 10070–10086.
46. T. Lu and Q. Chen, "Independent Gradient Model Based on Hirshfeld Partition: A New Method for Visual Study of Interactions in Chemical Systems," *Journal of Computational Chemistry* 43 (2022): 539–555.
47. T. Lu and F. Chen, "Multiwfn: A Multifunctional Wavefunction Analyzer," *Journal of Computational Chemistry* 33 (2012): 580–592.
48. T. Lu, "A Comprehensive Electron Wavefunction Analysis Toolbox for Chemists, Multiwfn," *Journal of Chemical Physics* 161 (2024): 082503.
49. S. Maeda, K. Ohno, and K. Morokuma, "Systematic Exploration of the Mechanism of Chemical Reactions: The Global Reaction Route Mapping (GRRM) Strategy Using the ADDF and AFIR Methods," *Physical Chemistry Chemical Physics* 15 (2013): 3683–3701.
50. S. Maeda, Y. Harabuchi, M. Takagi, et al., "Implementation and Performance of the Artificial Force Induced Reaction Method in the GRRM17 Program," *Journal of Computational Chemistry* 39 (2018): 233–251.
51. D. M. Anstine, R. Zubatyuk, and O. Isayev, "AIMNet2: A Neural Network Potential to Meet Your Neutral, Charged, Organic, and Elemental-Organic Needs," *Chemical Science* 16 (2025): 10228–10244.
52. B. M. Wood, M. Dzamba, X. Fu, et al., "UMA: A Family of Universal Models for Atoms," (2025), <https://arxiv.org/abs/2506.23971>.
53. S. Grimme, C. Bannwarth, and P. Shushkov, "A Robust and Accurate Tight-Binding Quantum Chemical Method for Structures, Vibrational Frequencies, and Noncovalent Interactions of Large Molecular Systems Parametrized for All Spd-Block Elements (Z = 1–86)," *Journal of Chemical Theory and Computation* 13 (2017): 1989–2009.
54. F. Neese, "The ORCA Program System," *WIREs Computational Molecular Science* 2 (2012): 73–78.
55. J.-D. Chai and M. Head-Gordon, "Long-Range Corrected Hybrid Density Functionals With Damped Atom-Atom Dispersion Corrections," *Physical Chemistry Chemical Physics* 10 (2008): 6615–6620.
56. J.-D. Chai and M. Head-Gordon, "Systematic Optimization of Long-Range Corrected Hybrid Density Functionals," *Journal of Chemical Physics* 128 (2008): 084106.
57. F. Weigend and R. Ahlrichs, "Balanced Basis Sets of Split Valence, Triple Zeta Valence and Quadruple Zeta Valence Quality for H to Rn: Design and Assessment of Accuracy," *Physical Chemistry Chemical Physics* 7 (2005): 3297–3305.
58. M. J. Frisch, G. W. Trucks, H. B. Schlegel, et al., *Gaussian 16, Revision C.01* (Gaussian, Inc., 2016).
59. L. W. Chung, W. M. C. Sameera, R. Ramozzi, et al., "The ONIOM Method and Its Applications," *Chemical Reviews* 115 (2015): 5678–5796.
60. K. Suzuki, K. Morokuma, and S. Maeda, "Multistructural Microiteration Technique for Geometry Optimization and Reaction Path Calculation in Large Systems," *Journal of Computational Chemistry* 38 (2017): 2213–2221.
61. K. Suzuki and S. Maeda, "Multistructural Microiteration Combined With QM/MM-ONIOM Electrostatic Embedding," *Physical Chemistry Chemical Physics* 24 (2022): 16762–16773.
62. K. Suzuki, "GRRM-OnioM," (2024), <https://github.com/Kimichi-Suzuki/GRRM-OnioM>.
63. C. Bannwarth, S. Ehlert, and S. Grimme, "GFN2-xTB—An Accurate and Broadly Parametrized Self-Consistent Tight-Binding Quantum Chemical Method With Multipole Electrostatics and Density-Dependent Dispersion Contributions," *Journal of Chemical Theory and Computation* 15 (2019): 1652–1671.
64. C. Bannwarth, E. Caldeweyher, S. Ehlert, et al., "Extended Tight-Binding Quantum Chemistry Methods," *WIREs Computational Molecular Science* 11 (2021): e1493.
65. S. Grimme, A. Hansen, S. Ehlert, and J.-M. Mewes, "r²SCAN-3c: A "Swiss Army Knife" Composite Electronic-Structure Method," *Journal of Chemical Physics* 154 (2021): 064103.
66. F. Neese, "Software Update: The ORCA Program System—Version 5.0," *WIREs Computational Molecular Science* 12 (2022): e1606.
67. N. Mardirossian and M. Head-Gordon, "ωB97X-V: A 10-Parameter, Range-Separated Hybrid, Generalized Gradient Approximation Density Functional With Nonlocal Correlation, Designed by a Survival-Of-The-Fittest Strategy," *Physical Chemistry Chemical Physics* 16 (2014): 9904–9924.
68. J. Zheng, X. Xu, and D. G. Truhlar, "Minimally Augmented Karlsruhe Basis Sets," *Theoretical Chemistry Accounts* 128 (2011): 295–305.
69. M. Cossi, N. Rega, G. Scalmani, and V. Barone, "Energies, Structures, and Electronic Properties of Molecules in Solution With the C-PCM Solvation Model," *Journal of Computational Chemistry* 24 (2003): 669–681.
70. E. F. Pettersen, T. D. Goddard, C. C. Huang, et al., "UCSF Chimera—A Visualization System for Exploratory Research and Analysis," *Journal of Computational Chemistry* 25 (2004): 1605–1612.
71. T. D. Goddard, C. C. Huang, E. C. Meng, et al., "UCSF ChimeraX: Meeting Modern Challenges in Visualization and Analysis," *Protein Science* 27 (2018): 14–25.
72. Blender Online Community, *Blender-A 3D Modelling and Rendering Package* (Stichting Blender Foundation, 2018).
73. S. Matsutani, "TStrail_TSlog_Files," (2026), <https://doi.org/10.19061/ichoem-bd-6-639>.

Supporting Information

Additional supporting information can be found online in the Supporting Information section. **Figure S1:** Example of agglomerative clustering. Here, 500 reactant structures were sampled and subjected to agglomerative clustering. Colors indicate the structural similarity of conformers, where conformers with the same color are more similar to each other than to those with different colors. **Figure S2:** Results from the initial and second stages of TStrail for three aldol reactions. (A) Target model aldol reaction yielding the (1S,2R) product. (B–D) TS conformers calculated in the initial and second stages of TStrail. Black circles represent TS conformers calculated in the initial stage, whereas orange triangles represent TS conformers calculated in the second stage of TStrail. **Table S1:** Distortion-interaction analysis calculated at the ωB97X-V/def2-TZVPP level of theory for the lowest- and highest-energy TS conformers of the reaction between the aldehyde and enol obtained in the initial stage of TStrail (Energy units in kcal/mol). **Table S2:** Distortion-interaction analysis calculated at the ωB97X-V/def2-TZVPP level of theory for the lowest- and highest-energy TS conformers of the reaction between the aldehyde and enolate anion obtained in the initial stage of TStrail (Energy units in kcal/mol). **Figure S3:** The result from

the initial and second stage of TStrail for BINOL-catalyzed Mannich-type reaction, where the TS conformers calculated in the initial stage of TStrail are represented as red circles and blue triangles, while the TS conformers calculated in the second stage of TStrail are represented as red pluses and blue crosses. **Figure S4:** Boltzmann distribution of TS conformers calculated at the second stage of TStrail. The red lines represent (S)-TS conformers while blue lines represent (R)-TS conformers. CPCM(Toluene)- ω B97X-V/def2-TZVPP// ω B97X-D/def2-SV(P) level of theory. **Figure S5:** Energy diagram of the IDPi-catalyzed intramolecular hydroamination reaction. The black and gray lines represent the structures obtained from the initial calculations: the black line corresponds to the stepwise pathway, whereas the gray dotted line corresponds to the concerted pathway. Both pathways lead to the (R)-product. The red and blue lines represent the lowest-energy TSs identified in the second stage of TStrail, with the red line corresponding to the lowest-energy (R)-TS and the blue line corresponding to the lowest-energy (S)-TS. The calculations were conducted using CPCM(Toluene)- ω B97X-V/def2-TZVPP//r²SCAN-3c level of theory. Free energies are in kcal/mol. **Figure S6:** ONIOM layer definition for the intramolecular hydroamination reaction. Atoms shown as colored spheres correspond to the high layer, whereas atoms shown in light gray as sticks correspond to the low layer. **Figure S7:** The result from the initial and second stage of TStrail for IDPi-catalyzed intramolecular hydroamination reaction, where the TS conformers calculated in the initial stage of TStrail are represented as red circles and blue triangles, while the TS conformers calculated in the second stage of TStrail are represented as green pluses and purple crosses. **Figure S8:** Boltzmann distribution of TS conformers calculated at the second stage of TStrail. The red lines represent (R)-TS conformers while blue lines represent (S)-TS conformers. Free energies were calculated at CPCM(Toluene)- ω B97X-V/def2-TZVPP//r²SCAN-3c level of theory. **Figure S9:** Superimposed (R)-TS conformers obtained in the initial stage of TStrail. The IDPi anion is shown as pink sticks, whereas the nosyl and phenyl groups of the reactant structures are highlighted as ball-and-stick models. **Figure S10:** 3D structures of the most stable (R)-TS and (S)-TS identified in the initial stage of TStrail. (A–C) Representations of the (R)-TS, with the IDPi catalyst shown in pink and the substrate shown in ball-and-stick representation. (D–F) Representations of the (S)-TS, with the IDPi catalyst shown in pink and the substrate shown in ball-and-stick representation. **Figure S11:** Energy diagram of three aldol reactions calculated in this work. The calculations were conducted using ω B97X-V/def2-TZVPP// ω B97X-D/def2-SV(P) level of theory. Thermal corrections have been performed at 298.15 K and 1 atm. **Figure S12:** Energy diagram of BINOL-catalyzed Mannich-type reaction calculated in this work. The calculations were conducted using CPCM(Toluene)- ω B97X-V/def2-TZVPP// ω B97X-D/def2-SV(P) level of theory. Thermal corrections have been performed at 195.15 K and 1 atm. **Figure S13:** Energy diagram of IDPi-catalyzed intramolecular hydroamination reaction calculated in this work. The calculations were conducted using CPCM(Toluene)- ω B97X-V/def2-TZVPP//r²SCAN-3c level of theory. Thermal corrections have been performed at 313.15 K and 1 atm.